

Task # 01 - Student Result Percentage

Ali obtained 75, 82, 68 marks in three subjects. Each subject has a total of 100 marks. Find obtained marks and Percentage

PAC

Given Data

Marks Ali scored in
3 subjects

Required Results

Marks obtained and Percentage

Pro

Processing Required

Sum of all = Sum of 3 marks

Total = 300

Percentage = $(\text{Sum} / \text{Total } 300) * 100$

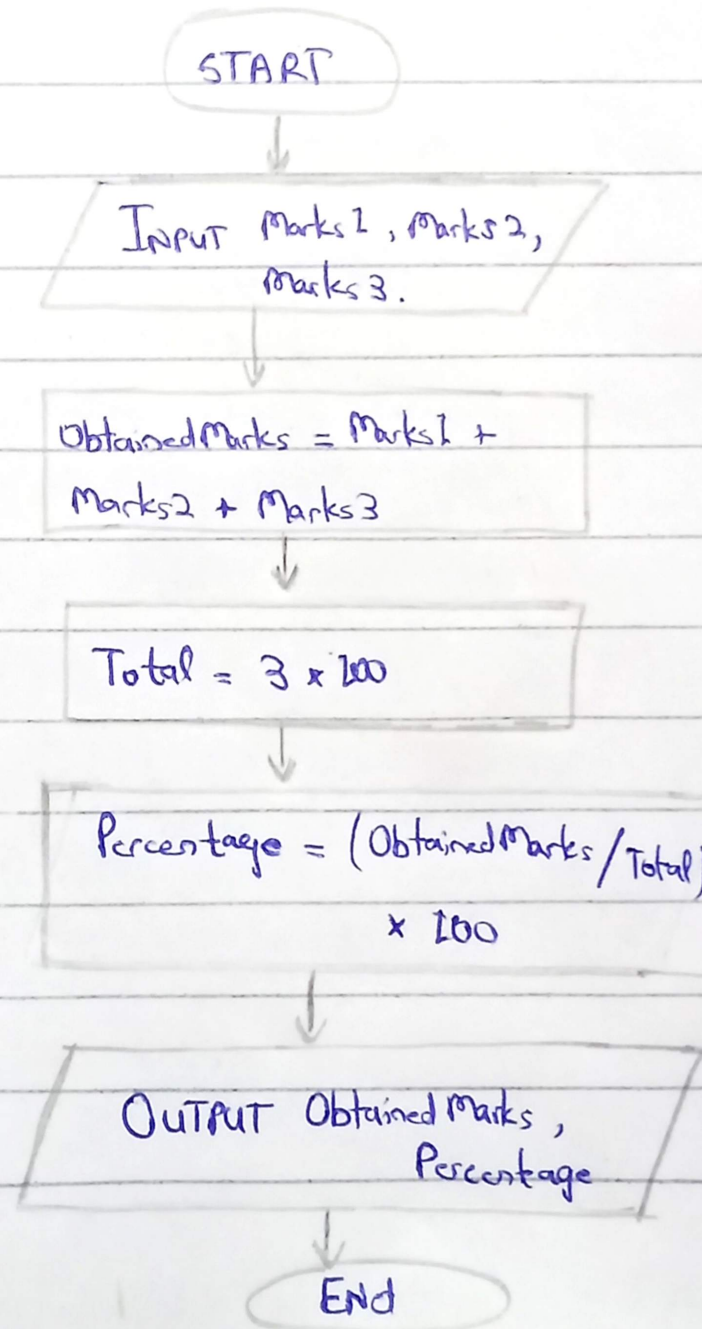
Solution Alternative

1) Enter the marks scored in 3 subjects

2) Add all of them to get obtained marks

3) To get Percentage divided obtained marks by total: i.e 300 and multiply by 100.

Flow Chart



Ipo

Input	Processing	Module Reference	Output
Sub1Marks	Obtained Marks = Sub1-	Sum	Obtained Marks
Sub2Marks	Marks + Sub2Marks + Sub3-	Sum	Percentage
Sub3Marks	Marks	Sum	
	Total = 100×3	Calculate	
	Percentage = (Obtained	Calculate	
	Marks / Total) $\times 100$	Calculate	

Task #02 - Shopping Discount

Sara wants to buy a school bag costing Rs. 2,000. The shop is offering a 15% discount.

Find Discounted Amount

PAC

Given Data

Original Price and
Discount Rate ~~given~~

Required Results

Discounted Amount

Processing Required

$$\text{Discount} = (\text{Original Price} \times \text{Discount Rate}) / 100$$

$$\text{Discounted Amount} = (\text{Original Price} - \text{Discount})$$

Solution Alternative

1) Input Original Price and discount Rate.

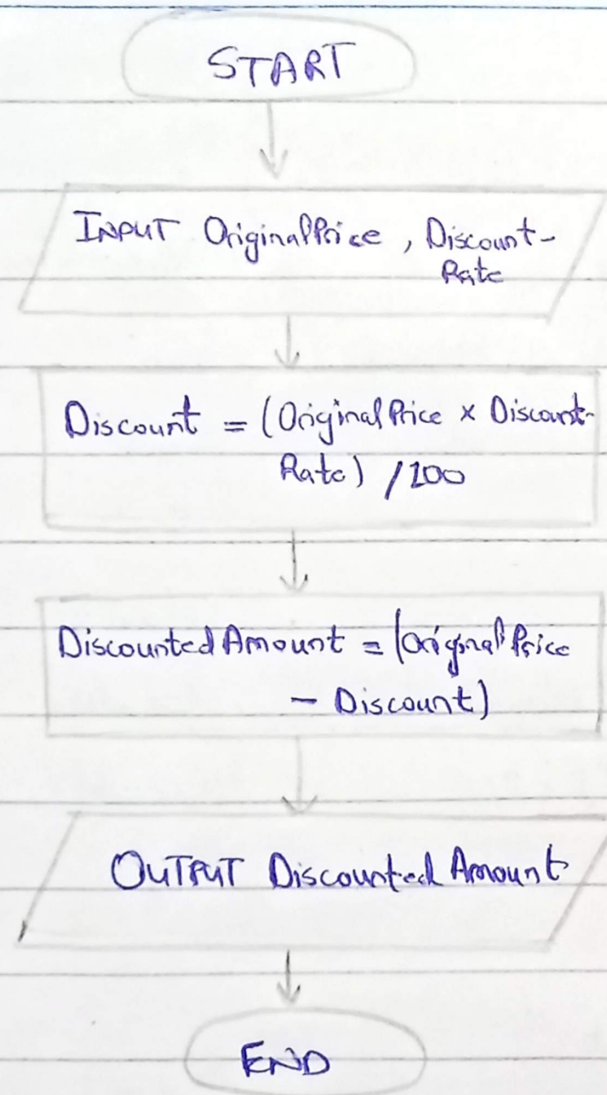
2) Calculate the discount by multiplying original price and discount rate and dividing them by 100

3) Now calculate Discounted Amount by subtracting ~~original price from~~ discount from original Price

IPO

Input	Processing	Module Reference	Output
Original Price Discounted Rate	③ $\text{Discount} = (\text{Original Price} \times \text{Discount Rate}) / 100$	Calculate	Discounted Amount
	④ $\text{Discounted Amount} = (\text{Original Price} - \text{Discount})$	Calculate	
	① Enter Original Price	Read	
	② Enter Discounted Rate	Read Read	

Flow Chart



Task# 03 - Water Tank Volume

A water tank is 20 meters long, 10 meters wide, and 5 meters high. Find volume

PAC

Given Data

Tank length, Tank width and
Tank height.

Required Results

Volume of Tank.

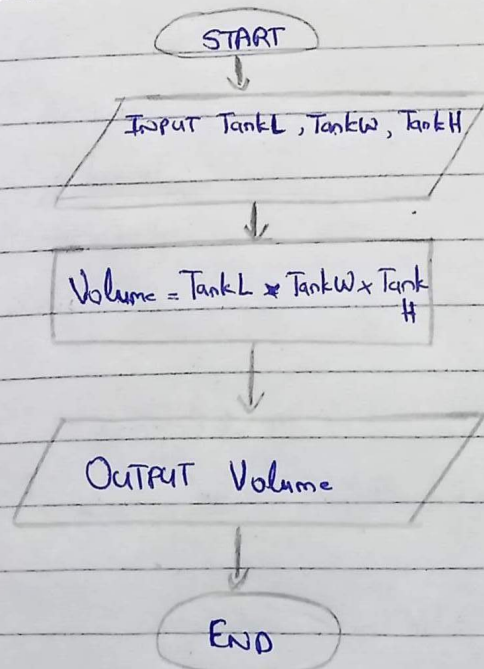
Processing Required

$$\text{Volume} = \text{Tank length} \times \text{Tank width} \times \text{Tank height}$$

Solution Alternative

- 1) Enter length, width, ~~height~~ and height of water tank
- 2) Calculate volume by multiplying every measurement ~~value~~ to get volume of water tank.

Flowchart



IPO

Input	Processing	Module Reference	Output
Tanklength	① Enter Tank length	Read	Volume
Tankwidth	② Enter Tank width	Read	
Tankheight	③ Enter Tank height	Read	
	④ $\text{Volume} = \text{Tanklength} \times \text{Tankwidth} \times \text{Tankheight}$	Calculate	

Task #04 - Rectangle Area

A rectangular classroom board has a length of 15m and a width of 8m. Find Rectangle Area

PAC

Given Data

length and width of
classroom board

Required Result

Rectangle Area

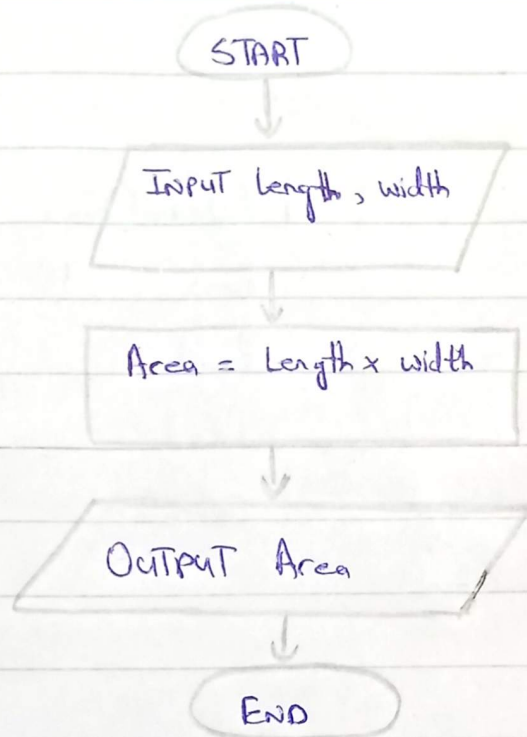
Processing Required

$$\text{Area} = \text{length} \times \text{width}$$

Solution Alternative

- 1) Enter length and width of the classroom board.
- 2) Calculate Area of board by multiplying the two values together.

Flowchart



Algorithm

Start

Input length // $L = 15 \text{ cm}$

Input width // $W = 8 \text{ cm}$

Area = length $\times$ width // Area = $L \times W$

Print Area // Output Area = 120 cm^2

Stop

Task #05 - University Field Trip

A university bus travels 12km north, 8km east, 10km south, and 5km west during a field trip. Find Total Distance required.

PAC

Given Data

Distances from
different paths, D_1, D_2, D_3, D_4

Required Result

Total Distance

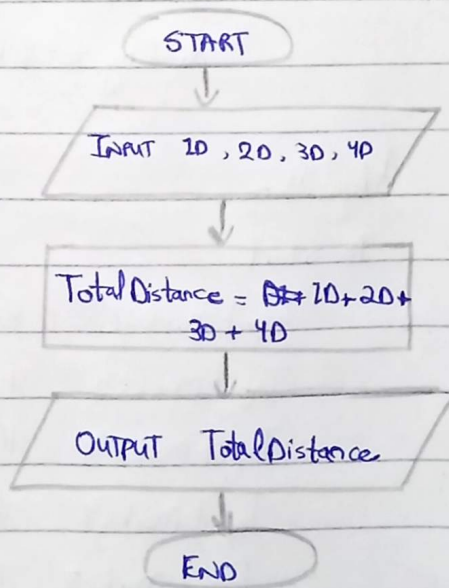
Processing Required

Total Distance = $D_1 + D_2 + D_3 + D_4$

Solution Alternatives

- 1) Enter all the different distances as D_1, D_2, D_3 etc
^{10, 20, 30}
- 2) Now add them all to get the value of Total distance that is covered.

Flow Chart



Algorithm

Start

Distance travelled in South

Input South Distance // 10km

Input North Distance // 12km

Input East Distance // 8km

Input West Distance // 5km

Total Distance = South Distance +

North Distance + West Distance +

East Distance // 35km

Print Total Distance // 35km

Stop

Task # 06 - Average Marks

A student received the marks in five subjects 80, 75, 90, 85, 70. Find Average Marks

PAC

Given Data

Marks in 5 subjects

Sub1, Sub2, Sub3, Sub4, Sub5

Required Results

Average Marks •

Processing Required

$$\text{Total} = \text{Sub1} + \text{Sub2} + \text{Sub3} + \text{Sub4} + \text{Sub5}$$

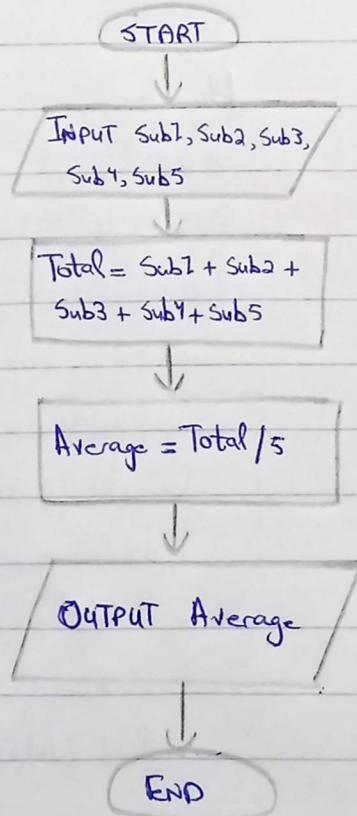
$$\text{No. of subjects} = 5$$

$$\text{Average} = \text{Total} / \text{No. of subjects}$$

Solution Alternatives

- 1) Enter marks for all 5 subjects
- 2) Calculate Total marks by adding them all
- 3) Calculate Average by ~~div~~ dividing total by No. of subject

Flow Chart



Algorithm

To start

Input Marks1 // 80

Input Marks2 // 75

Input Marks3 // 90

Input Marks4 // 85

Input Marks5 // 70

Total = Marks1 + Marks2 + Marks3 + Marks4 + Marks5 // 400

Average = Total / 5 // 80

~~Print~~ Print Average // 80

End